Summary of Math 312

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1 Induction

Some basic properties of the natural numbers:

- Mathematical induction

  Suppose $S \subseteq \mathbb{N}$ such that:

  - $1 \in S$
  - $\forall n \in \mathbb{N}, n \in S \implies n+1 \in S$

- Strong Induction

  Suppose $S \subseteq \mathbb{N}$ such that:

  - $1 \in S$
  - $\forall n \in \mathbb{N}, \forall k \in \mathbb{N}, 1 \leq k \leq n \implies k \in S \implies n+1 \in S$

- Well-Ordering Principle: Every non-empty subset of $\mathbb{N}$ has a least element.

Note that the above three are equivalent.

1.1 Examples

Let $\alpha = \frac{1+\sqrt{5}}{2}, \beta = \frac{1-\sqrt{5}}{2}$, Prove that the n-th Fibonacci number $F_n = \frac{\alpha^n - \beta^n}{\alpha - \beta} = \frac{\alpha^n - \beta^n}{\sqrt{5}}$.

Proof. Note that α, β are the roots of $x^2 - x - 1 = 0$. Then we have $\alpha^2 = \alpha + 1, \beta^2 = \beta + 1$. We will prove the statement by **Strong** induction on n.

Base Case: $n = 1, F_1 = 1 = \frac{\alpha-\beta}{\sqrt{5}} = \frac{\alpha^1-\beta^1}{\sqrt{5}}$.

$n = 2, F_2 = 1 = \frac{\alpha^2-\beta^2}{\sqrt{5}}$.

Inductive Step: Suppose $F_k = \frac{\alpha^k-\beta^k}{\sqrt{5}}$ for all $k \leq n$. Then we have $F_{n+1} = F_n + F_{n-1} = \frac{\alpha^n-\beta^n}{\sqrt{5}} + \frac{\alpha^{n-1}-\beta^{n-1}}{\sqrt{5}} = \frac{\alpha^{n-1}(\alpha+1)-\beta^{n-1}(\beta+1)}{\sqrt{5}} = \frac{\alpha^{n+1}-\beta^{n+1}}{\sqrt{5}}$.

Used the properties of α, β in the last step. □

Another example:

Prove that all positive integers are a product of primes. Here we treat the empty product as 1 and the product of one prime as itself.

Proof. We use proof by contradiction to show the use of the well-ordering principle.

Let S be the set of positive integers which cannot be written as products of primes. **Assume** that S is not empty:

Then by the well-ordering principle, S has a least element n. Since n is the least element, it cannot be prime.

Then $n = ab$ for some $a, b \in \mathbb{N}$. Then $a, b < n$. Since n is the least element, $a, b \notin S$. Then a, b can be written as products of primes. Then $n = ab$ can be written as a product of primes. This is a contradiction.

Therefore, S is empty and all positive integers can be written as products of primes. □

1.2 The division algorithm

Let $a \in \mathbb{Z}, a \geq 1$. $\forall n \in \mathbb{Z}, \exists q, r \in \mathbb{Z}$ such that $n = aq + r$ and $0 \leq r < a$. Moreover, q, r are unique.

Proof. Prove using the well-ordering principle.

Let $S = \{s \in \mathbb{Z} | s \geq 0 \wedge s = n - aq \wedge a, q, n \in \mathbb{Z}\}$. We claim that S is not empty.

Indeed $n - aq \geq 0 \iff n \geq aq \iff q \leq \frac{n}{a}$, given $a > 0$. So there are infinitely many q that satisfy the condition, which proves that S is not empty.

By the well-ordering principle, S has a least element r. Then $r = n - aq$ for some q. Then $n = aq + r$. It remains to show that $0 \leq r < a$. We argue by contradiction.

If $r \geq a$, then $r - a \geq 0$. Then $r - a = n - aq - a = n - a(q+1) \in S$. But $r - a < r$, which contradicts the fact that r is the least element of S. Therefore, $r < a$.

To show uniqueness, suppose $n = aq + r = aq' + r'$, where $0 \leq r, r' < a$. Our goal is to show that $q = q'$ and $r = r'$.

Subtracting the two equations, we have $a(q - q') = r' - r$. Notice that the LHS is divisible by a and the RHS is less than a but greater than 0. Therefore, $r' - r = 0$ and $q - q' = 0$. Therefore, $q = q'$ and $r = r'$.

This completes the proof. □

Remark: q stands for **quotient** and r stands for **remainder**.

Also, the Division algorithm is a **theorem**. The actual algorithm to find q and r is called **long division**.

2 Primes

Definition 1. *A positive integer p is called **prime** if $p \geq 2$ and the only positive divisors of p are 1 and p.*

2.1 Euclid's Theorem

Theorem 1. *There are infinitely many primes.*

Proof. Suppose there are only finitely many primes $p_1, p_2, \ldots, p_n$.

Let $N = p_1 p_2 \ldots p_n + 1$. (Products of primes plus 1)

Then N is not divisible by any of the primes $p_1, p_2, \ldots, p_n$. Therefore, N is either prime or divisible by a prime larger than p_n.

This is a contradiction. Therefore, there are infinitely many primes. □

The theorem can be used to generate infinitely many primes. For example, $2, 2 \cdot 3 + 1 = 7, 2 \cdot 3 \cdot 5 + 1 = 31, \ldots$ are all primes. However, this is not a good way to generate primes because the numbers get very large very quickly.

2.2 Sieve of Eratosthenes

Theorem 2. *Let $n \in \mathbb{N}, n \geq 2$. Then there exists a prime p such that $p \leq n$.*

The key idea is that if $n = ab$ and n is a composite, then at least one of a, b is less than or equal to $\sqrt{n}$. So we only need to check up to $\sqrt{n}$ to see if n is a prime.

For example, to check the primes within 25, we only need to check up to $\sqrt{25} = 5$. List out all the numbers from 2 to 25. Then cross out all the multiples of 2, 3, 5. The remaining numbers are all primes.

2.3 The prime number theorem

Theorem 3. *Let $\pi(x)$ be the number of primes less than or equal to x. Then $\lim_{x\to\infty} \frac{\pi(x)}{x/\ln x} = 1$.*

In other words, the number of primes less than or equal to x is approximately $\frac{x}{\ln x}$.

3 Division

3.1 Greatest common divisor

Definition 2. *Let $a, b \in \mathbb{Z}, (a, b) \neq (0, 0)$. The **greatest common divisor** of a and b, denoted by $\gcd(a, b)$, is the largest positive integer that divides both a and b.*

Simple properties of gcd:

- $a \neq 0 \implies \gcd(a, 0) = |a|$.
- $\gcd(a, b) \geq 1$.
- $\gcd(a, b) = d \implies \gcd(\frac{a}{d}, \frac{b}{d}) = 1$.

Proof.

- Common divisors of a and 0 are all divisors of a. The largest one is $|a|$.
- We know 1 is a common divisor of a and b. Therefore, $\gcd(a, b) \geq 1$.
- Suppose $d = gcd(a, b), e = gcd(\frac{a}{d}, \frac{b}{d})$.

  We write $\frac{a}{d} = ek, \frac{b}{d} = em$ for some $k, m \in \mathbb{Z}$. Then $a = (de)k, b = (de)m$. Now we can know that de is a common divisor of a and b. Thus $de \leq d$. Since $d \geq 1$, we have $e \leq 1$. Therefore, $e = 1$. Proof concluded.

$\square$

3.1.1 Some Crollaries

1. An integer e is a common divisor of a and b if and only if $e|gcd(a, b)$.

   Proof. If e is a common divisor of a and b, then $e|a$ and $e|b$. Therefore, $e|gcd(a, b) = ma + nb$ (Bezout's identity which will be proven below) for some $m, n \in \mathbb{Z}$. Therefore, $e|gcd(a, b)$.

   If $e|gcd(a, b)$, then $gcd(a, b) = ek$ for some $k \in \mathbb{Z}$. Then $gcd(a, b) = ek = ma + nb$. Therefore, $e|a$ and $e|b$. Therefore, e is a common divisor of a and b. □

2. Let c be an integer, then $ax + by = c$ has an integer solution if and only if $gcd(a, b)|c$.

   Proof.

   If $ax + by = c$ has an integer solution, then $gcd(a, b)|ax + by = c$.

   If $gcd(a, b)|c$, then $c = k(gcd(a, b)) = k(ax + by) = (ka)x + (kb)y$. Therefore, $ax + by = c$ has an integer solution. □

3.2 Bezout's identity

An integer linear combination of a and b is an integer of the form $ax + by$, where $x, y \in \mathbb{Z}$.

Theorem 4. *gcd(a, b) is the smallest positive integer that can be written as an integer linear combination of a and b.*

Proof. Let S be the set of positive integers that can be written as an integer linear combination of a and b. We want to show that $gcd(a, b)$ is the smallest element of S.

Note that since $(a, b) \neq (0, 0)$, S is not empty. By the well-ordering principle, S has a least element d. We want to show that $d = gcd(a, b)$. To show it, we need to show that:

- $gcd(a, b)|d$. This tells us that $gcd(a, b) \leq d$.
- $d|a$ and $d|b$. This tells us that $d \leq gcd(a, b)$.

By definition, $d = x_0a+y_0b$ for some $x_0, y_0 \in \mathbb{Z}$ which is a linear combination of a and b. But we know that $gcd(a, b)|(ma+nb), \forall m, n \in \mathbb{Z}$. Also, $d = x_0a+y_0b$. Therefore, $gcd(a, b)|d$.

Now we want to show that $d|a$ and $d|b$. Let $a = dq + r$ where q is the quotient and r is the remainder. Just need to show that r = 0. Note that:

$$r = a - dq = a - (x_0a + y_0b)q = a(1 - x_0q) + b(-y_0q)$$

So r is a linear combination of a and b. Since r is a remainder, $0 \leq r < d$. Since d is the smallest positive integer in S, r must be 0. Therefore, $d|a$. Similarly, $d|b$. □

3.3 Euclidean Division Algorithm

The Euclidean algorithm is a highly **recursive** and fast algorithm to find the gcd of two numbers.

The complexity of the algorithm is $O(\log n)$, where n is the larger of the two numbers.

Before starting, we need to know the following lemma:

3.3.1 Lemma

Let $a, b, q, r \in \mathbb{Z}$ such that $a = bq + r$. Then $gcd(a, b) = gcd(b, r)$.

Proof. Let $d = gcd(a, b), d' = gcd(b, r)$. We want to show that $d = d'$.

We know that $d'|b$ and $d'|r$. Therefore, $d'|bq + r = a$. Therefore, $d'|a$. So $d' \leq d$, it is a common divisor of a and b.

At the same time, $d|a$ and $d|b$. Therefore, $d|bq + r = a$. Therefore, $d|a$ and $d|r$. So $d \leq d'$, it is a common divisor of b and r.

Combine two results: $d = d'$. □

3.3.2 Euclidean algorithm

The algorithm applies the lemma repeatedly until we get $r = 0$ if we have $a, b \geq 0$. Then the last non-zero remainder is the gcd of a and b.

Give a recursion code example:

```
int gcd(int a, int b) {
    if (b == 0) return a;
    if (a == 0) return b;
```

```
        if (a < b) return gcd(b % a, a);
        return gcd(b, a % b);
    }
```

So basically, we keep replacing $\gcd(a, b)$ with $\gcd(b, r)$ until $r = 0$ if we choose $a \geq b$ the whole time. Then the last non-zero remainder is the gcd of a and b.

3.3.3 Number of steps in Euclidean algorithm

Lemma:

Let $a, b \in \mathbb{Z}, a \geq b \geq 1$. We let $a = bq + r$ where $0 \leq r < b$. Then $r < \frac{a}{2}$.

Proof.

- Case 1: $b \leq \frac{a}{2}$. Then $r < b \leq \frac{a}{2}$.
- Case 2: $b > \frac{a}{2}$. Then $r = a - bq < a - \frac{a}{2}q = \frac{a}{2}(2 - q) \leq \frac{a}{2}$. Given $q \geq 1$ since $a \geq b \leq 0 \wedge r \geq 0$.

□

Corollary:

Assume $a \geq b \geq 1$. Then the number of steps in the Euclidean algorithm is at most $2 \log_2 a$.

Proof. By the lemma above, we know that for each iteration, $r < \frac{a}{2}$. Therefore, after k iterations, $r < \frac{a}{2^k}$.

But r is a positive integer otherwise it terminates (The last r before becoming zero is the gcd, we have r = 0 to be the termination point in the sample code since it is easier to program).

We want to find the smallest k such that $\frac{a}{2^k} < 1$ (termination point). This is equivalent to $2^k > a$. Therefore, $k > \log_2 a$. Therefore, the number of iterations is at most $\log_2 a$. □

3.3.4 Lame's theorem

Theorem 5. *Suppose $a, b \in \mathbb{Z}, a \geq b \geq 1$. Let d be the number of steps in the Euclidean algorithm. Then $d \leq 5 \log_{10} a$.*

The proof can be found on Wikipedia. The $\log_{10}$ is used because the proof involved Fibonacci numbers for some reason.

Now compare the bound by Lame's theorem and the bound by the corollary above. We can compare the two bounds:

$$2\log_2 a = \frac{2}{\log_{10} 2}\log_{10} a \approx 6.64\log_{10} a > 5\log_{10} a$$

So Lame's theorem is a better bound than the corollary above.

3.4 Linear equations

Solve linear equations in the form of $ax+by=c$ where $a, b, c \in \mathbb{Z}$, $(a, b) = (0, 0)$. And we are interested in the integer solutions of x, y. Note that real solutions are easy to find and there are infinitely many of them.

Bezout's identity proved that an integer solution exists if and only if $\gcd(a, b)|c$. $\gcd(a, b)$ is the smallest positive integer that can be written as an integer linear combination of a and b. So all linear combinations of a and b are multiples of $\gcd(a, b)$.

Now we want to find all the integer solutions.

- Describe the set of all integer solutions.
- Develop an algorithm to find all the integer solutions if exist

Let $d = \gcd(a, b)$.

1. if $c = d$, by Bezout's theorem, there exists $ax + by = d$ for some $x, y \in \mathbb{Z}$. To find them, use the Euclidean algorithm. Set $r_0 = a, r_1 = b$ assuming $a \geq b \geq 0$. Then $r_0 = r_1q_1 + r_2$. Then $r_1 = r_2q_2 + r_3$. Keep going until $r_{n-1} = r_nq_n + 0$. Then $r_n = d$. Then we can find x, y by back substitution.

   Back substitution:

   - start from $d = r_{n+1} = r_{n-1} - r_nq_n = r_{n-1} - q_n(r_{n-2} - r_{n-1}q_{n-1})$ So moving from a linear combination of r_{n-1} and r_n to a linear combination of r_{n-2} and r_{n-1}.
   - continues until we get a linear combination of $r_0 = a$ and $r_1 = b$.

   For example: find an integer solution for $154x + 35y = 7$.

$$\begin{aligned} 154 &= 35 \cdot 4 + 14 \\ 35 &= 14 \cdot 2 + 7 \\ 14 &= 7 \cdot 2 + 0 \end{aligned}$$

So $\gcd(154, 35) = 7$. Then we can find x, y by back substitution.

$$\begin{aligned} 7 &= 35 - 14 \cdot 2 \\ &= 35 - (154 - 35 \cdot 4) \cdot 2 \\ &= 35 \cdot 9 - 154 \cdot 2 \end{aligned}$$

So $x = -2, y = 9$ is a solution.

Another example: $553x + 327y = 1$.

$$\begin{aligned} 553 &= 327 \cdot 1 + 226 \\ 327 &= 226 \cdot 1 + 101 \\ 226 &= 101 \cdot 2 + 24 \\ 101 &= 24 \cdot 4 + 5 \\ 24 &= 5 \cdot 4 + 4 \\ 5 &= 4 \cdot 1 + 1 \\ 4 &= 1 \cdot 4 + 0 \end{aligned}$$

So $\gcd(553, 327) = 1$. Then we can find x, y by back substitution.

$$\begin{aligned} 1 &= 5 - 4 \\ &= 5 - (24 - 5 \cdot 4) = 5 \cdot 5 - 24 \cdot 1 \\ &= (101 - 24 \cdot 4) \cdot 5 - 24 \cdot 1 = 101 \cdot 5 - 24 \cdot 21 \\ &= 101 \cdot 5 - (226 - 101 \cdot 2) \cdot 21 = 101 \cdot 47 - 226 \cdot 21 \\ &= (327 - 226) \cdot 47 - 226 \cdot 21 = 327 \cdot 47 - 226 \cdot 68 \\ &= 327 \cdot 47 - (553 - 327 \cdot 1) \cdot 68 = 327 \cdot 115 - 553 \cdot 68 \end{aligned}$$

Now we can find $x = -68, y = 115$.

2. Back to $ax + by = c$. Now assume that $d = \gcd(a, b) | c$. So $c = td$, for some $t \in \mathbb{Z}$

   - In this case, find a solution (x_0, y_0) for $ax + by = d$.
   - Then $x = x_0 t, y = y_0 t$ is a solution for $ax + by = c$.
   - This is because $a(x_0 t) + b(y_0 t) = td = c$.

3. If $d \nmid c$, then there are no integer solutions.

   For example, $154x + 35y = 24$ has no integer solutions. This is because 24 is not a multiple of 7. t is not an integer.

4. Now we want to describe all solutions:

   Theorem 6. *Suppose* $a, b, c \in \mathbb{Z}, (a, b) \neq (0, 0)$.

   Let $d = \gcd(a, b)$. *Then the set of all integer solutions of* $ax + by = c$ *is given by*

   $$\{(x_0 + \frac{b}{d}t, y_0 - \frac{a}{d}t) | t \in \mathbb{Z}\}$$

   if $d|c \iff c = td$. *Otherwise, the set of all integer solutions is empty.*

   Note that (x_0, y_0) is a particular solution of $ax + by = c$. Here "general solution" means that (x, y) is a solution for every integer value of t, and every integer solution can be expressed in this form.

3.4.1 Key Lemma

Before proving the theorem, we need to prove the following lemma:

Key Lemma: Let $a, b, c \in \mathbb{Z}, (a, b) \neq (0, 0)$. Let $d = \gcd(a, b) = 1$. Then $a|bc \implies a|c$.

Note that the lemma is not true if $d \neq 1$.

Proof. By Bezout's theorem, $\gcd(a, b) = 1 \implies ax + by = 1$ for some $x, y \in \mathbb{Z}$. Then multiply both sides by c: $acx + bcy = c$. Then $a|bc \implies a|acx + bcy = c$ as defined in the Lemma. □

3.4.2 Proof of the general solution theorem

With the Lemma in hand, we can start proving the theorem:

Proof.

- First need to show that $x = x_0 + \frac{b}{d}t, y = y_0 - \frac{a}{d}t$ is a solution of $ax + by = c, \forall t \in \mathbb{Z}$.

  We have $ax_0 + by_0 = ax + by = c$. Then

  $$\begin{aligned} ax + by &= a(x_0 + \frac{b}{d}t) + b(y_0 - \frac{a}{d}t) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = c \end{aligned}$$

Therefore, we showed that $x = x_0 + \frac{b}{d}t, y = y_0 - \frac{a}{d}t$ is a solution of $ax + by = c, \forall t \in \mathbb{Z}$.

- Now we need to show that every integer solution can be expressed in this form.

  Subtract $ax_0 + by_0 = ax + by = c$ togeather:

  $$a(x - x_0) + b(y - y_0) = 0 \implies \frac{a}{d}(x - x_0) = -\frac{b}{d}(y - y_0)$$

  Note that $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$. Therefore, $\frac{a}{d}|(y - y_0)$ by the Key Lemma.

  Then let $y - y_0 = -\frac{a}{d}t$ for some $t \in \mathbb{Z}$. Substitute it into the equation:

  $$\frac{a}{d}(x - x_0) = \frac{b}{d}\frac{a}{d}t$$

  Here we have two cases:

  - If $a = 0$, then $by = c$. x can be arbitrary and $y = \frac{c}{b}$. Then $a(x_0 + \frac{b}{d}t) + b(y_0 - \frac{a}{d}t) = by_o = ax + by = c, \forall t \in \mathbb{R}$. Note that $b \neq 0$ in this case because $(a, b) \neq (0, 0)$. $a \neq 0 \wedge b = 0$ case is similar.
  - Now we talk about the general case. Divide both sides by $\frac{a}{d}$:

    $$x - x_0 = \frac{b}{d}t$$

    Then $x - x_0 = \frac{b}{d}t$ and we already have $y = y_0 - \frac{a}{d}t$. We have shown that every integer solution can be expressed in this form.

□

3.5 Fundedamental theorem of arithmetic

Theorem 7. *$\forall n \in \mathbb{N}, n \geq 1$, n can be written as a product of primes in a unique way.*

Note that one prime can appear multiple times in the product. For example, $12 = 2 \cdot 2 \cdot 3$. Also, we assume that $n = 1$ is an empty product which equals 1.

Usually, we collect the same primes together and write as

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$$

Now $p_1, p_2, \ldots, p_k$ are all distinct primes that appear in the product and $e_1, e_2, \ldots, e_k$ are the powers of the primes $\in \mathbb{N}$.

We have already proved the existence of the prime factorization in the previous section using the Well-Ordering Principle. Now we need to prove the uniqueness.

3.5.1 Corollary

Let p be a prime, $a_1, a_2, \ldots, a_n \in \mathbb{Z}$. Then $p|a_1a_2\ldots a_n \implies p|a_i$ for some i.

Proof. We prove by induction on n.

Base Case: $n = 1$. Then $p|a_1$. Obviously true.

Inductive Step: Suppose the statement is true for $n = k$. We want to show that the statement is true for $n = k + 1$.

Suppose $p|a_1a_2\ldots a_{k+1}$. Then $p|a_1a_2\ldots a_k$ or $p|a_{k+1}$. By the inductive hypothesis, $p|a_i$ for some i. □

3.5.2 Proof of the theorem

Proof. Let $n = p_1, p_2, \ldots, p_r = q_1, q_2, \ldots, q_s$ be two prime factorizations of n. We want to show that $r = s$ and $p_i = q_i$ for all i.

Cancel both primes on both sides if the same primes appear in both factorizations. Then we have new primes on both sides that do not equal each other

$$p_1p_2\cdots p_a = q_1q_2\cdots q_b$$

Our goal now is to show that there is nothing left on both sides.

Now by the corollary, $p_1|q_1q_2\ldots q_b$. Then $p_1|q_i$ for some i. But q_i is a prime. Therefore, $p_1 = q_i$. Then we can cancel p_1 and q_i from both sides. So this is a contradiction. Therefore, $r = s$ and $p_i = q_i$ for all i. We have proved the uniqueness. □

Remark: Uniqueness of prime factorization fails in many number systems other than the integers. For example: "even integers": In this system, then 6, 10, 30 and 50 are primes in this system. Then 300 = 6 · 50 = 10 · 30. So the prime factorization is not unique in this system.

3.5.3 Applications

Proposition: Suppose $n = p_1^{d_1} \times \cdots \times p_k^{d_k}, m = p_1^{e_1} \times \cdots \times p_k^{e_k}$ where $p_1, \ldots, p_k$ are distinct primes and $d_1, \ldots, d_k, e_1, \ldots, e_k \in \mathbb{N} \cup 0$. Then $\gcd(n, m) = p_1^{\min(d_1,e_1)} \times \cdots \times p_k^{\min(d_k,e_k)}$ and the largest common multiple $lcm(n, m) = p_1^{\max(d_1,e_1)} \times \cdots \times p_k^{\max(d_k,e_k)}$.

For example:

$$\begin{aligned} 35 &= 5 \times 7 & &= 2^0 \times 5^1 \times 7^1 \times 11^0 \\ 154 &= 2 \times 7 \times 11 & &= 2^1 \times 5^0 \times 7^1 \times 11^1 \\ \gcd(35, 154) &= 2^0 \times 5^0 \times 7^1 \times 11^0 & &= 7 \end{aligned}$$

In general, prime factorization is difficult for large numbers. The Euclidean algorithm is a much faster way to find the gcd of two numbers.

To show the Proposition is true, we need the following corollary:

Let us write the highest power of p appearing in the prime factorization of n as $v_p(n)$. For example: $v_2(12) = 2, v_3(12) = 1, v_5(12) = 0$. Then we have the following corollary:

If m and n are positive integers, then $m|n \iff v_p(m) \leq v_p(n)$ for all primes p.

Proof. Prove both directions:

- $\implies$: Suppose $m|n$. Then $n = mk$ for some $k \in \mathbb{Z}$. Then $v_p(n) = v_p(mk) = v_p(m) + v_p(k) \geq v_p(m)$.

- $\impliedby$: Suppose $0 \leq v_p(m) \leq v_p(n)$ for all primes p. Then $n = p_1^{v_{p_1}(n)} \times \cdots \times p_k^{v_{p_k}(n)} = m(p_1^{v_{p_1}(n)-v_{p_1}(m)} \times \cdots \times p_k^{v_{p_k}(n)-v_{p_k}(m)})$. Since $0 \leq v_p(m) \leq v_p(n)$, we make sure that $\forall i, v_{p_i}(n) - v_{p_i}(m) \in \mathbb{N} \cup 0$. Therefore, $m|n$.

$\square$

Now back to the Proposition:

Proof. By the Corollary above: $\forall$ common divisor m and n is of the form $p_1^{f_1} \ldots p_k^{f_k}$ where $f_i \leq \min(d_i, e_i)$.

Therefore, The greatest common divisor is exactly where $f_i = \min(d_i, e_i)$. Therefore, $\gcd(n, m) = p_1^{\min(d_1,e_1)} \ldots p_k^{\min(d_k,e_k)}$.

If we count for negatives, just add $\pm$ in front of the primes.

Proof of lcm is similar. $\forall$ lcm of m and n is of the form $p_1^{f_1} \ldots p_k^{f_k}$ where $f_i \geq \max(d_i, e_i)$. Therefore, lcm$(n, m) = p_1^{\max(d_1,e_1)} \ldots p_k^{\max(d_k,e_k)}$. $\square$

Corollary: $\gcd(n, m) \times \text{lcm}(n, m) = nm$.

Proof. $\max(d_i, e_i) + \min(d_i, e_i) = d_i + e_i$. Therefore, $\gcd(n, m) \times \text{lcm}(n, m) = nm$. $\square$

4 Congruences

Definition 3. *Let $a, b, n \in \mathbb{Z}, n \geq 2$. We say that a is **congruent** to b modulo n if $n|(a - b)$. We write $a \equiv b \pmod{n}$.*

Note that $\forall a, n \in \mathbb{Z}, n \geq 2, a \equiv b \pmod{n}$ for some $b \in \mathbb{Z}, 0 \leq b \leq n - 1$.

Congruences mod n may be thought of as a way of focusing on the remainder and casting out (ignoring) the quotient (multiples of n).

4.1 Congruence Class

Definition 4. *Let $n \in \mathbb{Z}, n \geq 2$. The **congruence class** of a modulo n is the set of all integers that are congruent to a modulo n. We write $[a]_n$.*

For example:

$$
\begin{aligned}
[0]_5 &= \{\ldots, -10, -5, 0, 5, 10, \ldots\} \\
[1]_5 &= \{\ldots, -9, -4, 1, 6, 11, \ldots\} \\
[2]_5 &= \{\ldots, -8, -3, 2, 7, 12, \ldots\} \\
[3]_5 &= \{\ldots, -7, -2, 3, 8, 13, \ldots\} \\
[4]_5 &= \{\ldots, -6, -1, 4, 9, 14, \ldots\}
\end{aligned}
$$

Note that $[a]_n = [b]_n \iff a \equiv b \pmod{n}$.

4.2 Properties of congruences

$$
\begin{aligned}
&a \equiv a \pmod{n} \forall a, n \in \mathbb{Z} \\
&a \equiv b \pmod{n} \implies b \equiv a \pmod{n} \\
&a \equiv b \pmod{n} \wedge b \equiv c \pmod{n} \implies a \equiv c \pmod{n} \\
&a \equiv b \pmod{n} \wedge c \equiv d \pmod{n} \implies a + c \equiv b + d \pmod{n} \\
&a \equiv b \pmod{n} \wedge c \equiv d \pmod{n} \implies ac \equiv bd \pmod{n} \\
&a \equiv b \pmod{n} \implies ac \equiv bc \pmod{n} \\
&a \equiv b \pmod{n} \implies a^k \equiv b^k \pmod{n} \forall k \in \mathbb{N}
\end{aligned}
$$

Note that it is reflective, symmetric and transitive. It is also closed under addition, multiplication and exponentiation.

4.2.1 Arithmetics on congruence classes

Arithmetics between congruence classes:

$$
\begin{aligned}
[a]_n + [b]_n &= [a + b]_n \\
[a]_n \times [b]_n &= [a \times b]_n \\
[a]_n^k &= [a^k]_n
\end{aligned}
$$

If $x_1 \equiv x_2 \pmod{n}, y_1 \equiv y_2 \pmod{n}$, then

- $x_1 + y_1 \equiv x_2 + y_2 \pmod{n}$
- $x_1 y_1 \equiv x_2 y_2 \pmod{n}$
- $x_1 - y_1 \equiv x_2 - y_2 \pmod{n}$

Proof. Suppose $x_1 - x_2 = an, y_1 - y_2 = bn$ for some $a, b \in \mathbb{Z}$. Then

- $(x_1 + y_1) - (x_2 + y_2) = an + bn = (a + b)n \equiv 0 \pmod{n}$ Thus $x_1 + y_1 \equiv x_2 + y_2 \pmod{n}$
- Subtraction is similar to addition proof
- $x_1 = x_2 + an, y_1 = y_2 + bn$ Then $x_1 y_1 = x_2 y_2 + n(ay_2 + bx_2 + ab) \equiv x_2 y_2 \pmod{n}$

$\square$

Using the rules above, we can calculate some insanely big numbers:

$$\begin{aligned} 48^{100} + 15 \times 70002 \pmod 7 &\equiv (-1)^{100} + 1 \times 2 \pmod 7 \\ &\equiv 1 + 2 \pmod 7 \\ &\equiv 3 \pmod 7 \end{aligned}$$

Note that if $d \equiv e \pmod n$, does not imply $a^d \equiv a^e \pmod n$. For example, $2 \equiv 9 \pmod 7$, but $2^2 \not\equiv 2^9 \pmod 7$.

4.2.2 Modular exponentiation

Modular exponentiation like raising an integer to a high power modulo n, is used in cryptography.

For large $x, y \equiv a^x \pmod n$ is easy to compute. On the other hand, given y, recovering x is hard. Without additional information, it is a "Discrete Logarithm Problem".

A quick algorithm for computing $a^x \pmod n$ is called "the method of repeated squaring". For example, $a^{13} \pmod n$ can be computed as follows:

For example: find $7^51 \pmod 1)7$

$$\begin{aligned} 7^1 &\equiv 7 \pmod{17} \\ 7^2 &\equiv 7 \times 7 \equiv 49 \equiv 15 \equiv -2 \pmod{17} \\ 7^4 &\equiv (-2)^2 \equiv 4 \pmod{17} \\ 7^8 &\equiv 4^2 \equiv 16 \equiv -1 \pmod{17} \\ 7^{16} &\equiv (-1)^2 \equiv 1 \pmod{17} \\ 7^{32} &\equiv 1^2 \equiv 1 \pmod{17} \\ 7^{51} &\equiv 7^{32} \times 7^{16} \times 7^2 \times 7^1 \equiv 1 \times 1 \times (-2) \times 7 \equiv -14 \equiv 3 \pmod{17} \end{aligned}$$

4.2.3 Representations of integers

Theorem 8. *Let $b \in \mathbb{Z}, b \geq 2$. Then $\forall m \in \mathbb{N}$ can be written as*

$$m = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$$

where $k \in \mathbb{N}, a_0, a_1, \ldots, a_k \in \mathbb{Z}, 0 \leq a_i \leq b - 1$ for all i.
Moreover, this representation is unique.

The theorem gives a representation of m in base b. a_k is the highest digit and a_0 is the lowest digit. If $b = 10$, then it is the normal decimal representation. We just simply leave out the subscript $b = 10$.

For example, $51 = 32 + 16 + 2 + 1 = 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 110011_2$.

Proof. Prove by strong induction on n.

Base Case: $b^0 \leq n < b^1$. Then $n = (n)_b$.

Inductive Step: Assume $n \geq b$ and the existence part of the theorem holds for any integer between 1 and $n - 1$. We want to show that the existence part of the theorem holds for n.

Divide n by b: $n = bq + r$ where $0 \leq r < b$. Then $0 \leq q < n$. By the inductive hypothesis, q can be written as

$$q = c_s b^s + c_{s-1} b^{s-1} + \cdots + c_1 b + c_0$$

Now $n = qb + r = c_s b^{s+1} + c_{s-1} b^s + \cdots + c_1 b^2 + c_0 b + r$. Therefore, n can be written as

$$n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$$

This is the representation of n in base b.

For uniqueness, again by induction:

Base Case: $n = b^0 \leq n < b^1$. Then $n = (n)_b$. It only has 1 digit and it has to be n, thus unique.

Inductive Step: Assume $n \geq b$ and the uniqueness part of the theorem holds for any integer between 1 and $n - 1$. We want to show that the uniqueness part of the theorem holds for n.

assume n can be expressed in two ways:

$$n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$$
$$n = c_s b^s + c_{s-1} b^{s-1} + \cdots + c_1 b + c_0$$

Then

$$a_0 = n \pmod b = c_0 \wedge 0 \leq a_0, c_0 \leq b - 1$$
$$\frac{1}{b}(n - a_0) = a_k b^{k-1} + \cdots + a_2 b + a_1 = \frac{1}{b}(n - c_0) = c_s b^{s-1} + \cdots + c_2 b + c_1$$

By induction assumption, $k - 1 = m - 1 \wedge a_i = c_i$ for all i. Therefore, $k = s \wedge a_i = c_i$ for all i. $\square$

4.2.4 Exponentiation arithmetic

Back to repeated squaring, the goal is to compute $a^x \pmod{n}$.

Pre-compute: $a, a^2, a^4, a^8, \ldots, a^{2^k} \pmod{n}$

Now write x in base 2: $x = 2^{k_1} + 2^{k_2} + \cdots + 2^{k_m}$ where $k_1 > k_2 > \cdots > k_m \geq 0$.

Then $a^x = a^{2^{k_1}} \times a^{2^{k_2}} \times \cdots \times a^{2^{k_m}} \pmod{n}$.

For example: Compute the last two digits of 53^{29}. Equivalently, it is $53^{29} \pmod{100}$.

Start with 29 in base 2: $29 = 16+8+4+1$. Then $53^{29} = 53^{16} \times 53^8 \times 53^4 \times 53^1 \pmod{100}$.

$$
\begin{aligned}
53^1 &= 53 \pmod{100} \\
53^2 &= 2809 \equiv 9 \pmod{100} \\
53^4 &= (53^2)^2 \equiv 81 \pmod{100} \\
53^8 &\equiv 81^2 = 6561 \equiv 61 \pmod{100} \\
53^{16} &\equiv 61^2 = 3721 \equiv 21 \pmod{100} \\
53^{29} &= 21 \times 61 \times 81 \times 53 \equiv 13 \pmod{100}
\end{aligned}
$$

4.3 Linear congruences

$ax \equiv b \pmod{n}$ is a linear congruence. We want to find all solutions of x given a, b, n.

The case where $b = 1$ is of particular interest. Here x is called the inverse of a modulo n. We write $x = a^{-1} \pmod{n}$.

Note that $ax \equiv b \pmod{n} \iff ax = b + nk$ for some $k \in \mathbb{Z}$. For example, $ax - nk = b$ for some $k \in \mathbb{Z}$. We know that $ax - nk = \gcd(a, n)$ by Bezout's theorem. Therefore, $ax - nk = \gcd(a, n)$ for some $k \in \mathbb{Z}$.

If it exists, we are interested in how many there are.

4.3.1 Solving linear congruences

Two solutions x_1 and x_2 are considered the same $\iff \exists y \in \mathbb{Z}$ s.t $ax - b = ny$ or Equivalently $ax - ny = b$.

In particular, if $\gcd(a, n)$ does not divide b, then the congruence $ax \equiv b \pmod{n}$ has no integer solution.

Theorem 9. *If $d = \gcd(a, n)|b$, then the congurence $ax \equiv b \pmod{n}$ has exactly d solutions*

Proof. Assume that $d = \gcd(a, n)|b$. Then the linear Diophantine equation $ax - b = ny \iff ax - ny = b$ for some $x, y \in \mathbb{Z}$ has a particular solution (x_0, y_0) by Bezout's theorem.

General solution: $x = x_0 + \frac{n}{d}t, y = y_0 - \frac{a}{d}t$ for some $t \in \mathbb{Z}$.

We are only interested in x here and up to multiples of n. In other words, $x_0 + \frac{n}{d}t_1$ and $x_0 + \frac{n}{d}t_2$ are considered the same if and only if $\frac{n}{d}t_1 \equiv \frac{n}{d}t_2 \pmod{n}$ or equivalently, $\frac{n}{d}(t_1 - t_2) \equiv 0 \pmod{n}$. (So that the congruence is the same)

$\frac{n}{d}(t_1 - t_2) \equiv 0 \pmod{n} \implies \frac{n}{d}(t_1 - t_2) = ns \implies t_1 - t_2 = ds$ for some $s \in \mathbb{Z}$. Therefore, $t_1 - t_2 \equiv 0 \pmod{d}$. Thus we get exactly d solutions by letting t range from 0 to $d - 1$. □

Conclusion: Every solution is $ax \equiv b \pmod{n}$ is of the form $x = x_0 + \frac{n}{d}t$ for some $t \in \mathbb{Z}$ where $d = \gcd(a, n)$ and x_0 is a particular solution. Two solutions $x_1 = x_0 + \frac{n}{d}t_1$ and $x_2 = x_0 + \frac{n}{d}t_2$ are considered the same if and only if $\frac{n}{d}(t_1 - t_2) \equiv 0 \pmod{n} \iff t_1 - t_2 \equiv 0 \pmod{d}$. Thus we get exactly d solutions by letting t range from 0 to $d - 1$.

4.3.2 Examples

For example, solve $10x \equiv 3 \pmod{12}$. Here $d = \gcd(10, 12) = 2$. Since $2 \nmid 3$, there are no solutions. The gcd needs to divide the congruence value.

Another example, $10x \equiv 4 \pmod{12}$. Here we expect 2 solutions. To find them, we start by finding a particular solution for $10x - 12y = 4$. Since the number is small, we can just guess the solution:

$x_0 = 4, y_0 = 3$ is the particular solution. Then the general solution is $x = 4 + \frac{12}{2}t = 4 + 6t$ for some $t \in \mathbb{Z}$.

- t ranges from 0 to 1.
- When $t = 0$, $x = 4$.
- When $t = 1$, $x = 10$.

So the solutions are $x \equiv 4 \pmod{12}$ and $x \equiv 10 \pmod{12}$.

4.3.3 Multuplicative inverses

Particular interesting case when $b = 1$. Then $ax \equiv 1 \pmod{n}$ is called the inverse of a modulo n. We write $x = a^{-1} \pmod{n}$.

In this case, the congruence has a solution if and only if $\gcd(a, n) = 1$, which also means the congruence has exactly one solution.

Not all congruence classes have an inverse. For example, $ax \equiv 1 \pmod{10}$:

$$\begin{aligned} 1 \times 1 &\equiv 1 \pmod{10} \\ 3 \times 7 &\equiv 1 \pmod{10} \\ 7 \times 3 &\equiv 1 \pmod{10} \\ 9 \times 9 &\equiv 1 \pmod{10} \end{aligned}$$

And the rest does not have one. It is because the rest have a gcd with 10 that is greater than 1.

Corollary: Let p be a prime, then every $a \not\equiv 0 \pmod{p}$ has an inverse modulo p. It is because $\gcd(a, p) = 1$.

Corollary: Let p be a prime, then $a \equiv a^{-1} \pmod{p} \iff a \equiv \pm 1 \pmod{p}$.

Proof. Indeed multiplying both sides by a, we get $a \equiv a^{-1} \pmod{p} \implies a^2 \equiv 1 \pmod{p} \iff (a-1)(a+1) \equiv 0 \pmod{p} \iff a \equiv \pm 1 \pmod{p}$.

If $a \not\equiv 1$ then $a - 1$ has a multiplicative inverse modulo p. Denote this multiplicative inverse by b. Time both side by b Then $b(a-1)(a+1) \equiv 0 \pmod{p} \implies a + 1 \equiv 0 \pmod{p} \implies a \equiv -1 \pmod{p}$. Therefore, $a \equiv b \pmod{p}$.

Similarly we can find that $a \not\equiv -1 \implies a \equiv 1 \pmod{p}$. So we proved the corollary. $\square$

For example, solve $17x \equiv 1 \pmod{5}5$. First, find the gcd of 17 and 55:

$$\begin{aligned} 55 &= 3 \times 17 + 4 \\ 17 &= 4 \times 4 + 1 \\ 4 &= 4 \times 1 \end{aligned}$$

Now use back substitution:

$$\begin{aligned} 1 &= 17 - 4 \times 4 \\ &= 17 - 4 \times (55 - 3 \times 17) \\ &= 13 \times 17 - 4 \times 55 \end{aligned}$$

Therefore, $13 \times 17 \equiv 1 \pmod{55}$. So the solution is $x \equiv 13 \pmod{55}$.